OFFICIAL TASK ASSIGNMENT & CURRICULUM SPECIFICATIONS

Internship Tasks & Instructions — CodeNova

Comprehensive technical handbook for interns enrolled in the **AI/ML** track. Contains detailed project task specifications, evaluation guidelines, milestone schedules, and submission protocols.

MODE & FORMAT	PROGRAM DURATION	PROGRAM FEES	SUBMISSION DEADLINE
Remote / Online	**4 Weeks (Self-Paced)**	**100% Free** ✓ Verified	**06 November 2026**

Internship Overview — AI/ML

The Artificial Intelligence and Machine Learning track immerses interns in empirical data analysis, predictive modeling, natural language processing, and grounded question-answering systems. Interns learn end-to-end ML workflows: exploratory data analysis, data cleaning, feature engineering, cross-validation, hyperparameter tuning, model evaluation metrics (Precision, Recall, F1, RMSE), and ethical AI deployment without hallucinations.

Internship Perks & Benefits

- ✔ **Internship Offer Letter** — Official verification letter confirming selection into Batch 01.
- ✔ **Certificate of Completion** — Accredited credential subject to completion criteria.
- ✔ **Letter of Recommendation** — Official LOR awarded based on exceptional performance.
- ✔ **Real-World Project Experience** — Practical, resume-ready portfolio software implementations.
- ✔ **Hackathons & Achievements** — Opportunity to participate in CodeNova hackathons & unlock achievements.
- ✔ **Community & Collaboration** — Peer-to-peer collaboration and networking with mentors.

General Instructions for Interns

1. Share an internship update on LinkedIn and tag the official **CodeNova** page.
2. Complete the assigned tasks within the internship timeline (by 06 November 2026).
3. Upload source code to GitHub using a clearly named repository for each completed project.
4. Add a descriptive README.md explaining project features, tech stack, setup, and screenshots.
5. Provide a working demo link when applicable (e.g., hosted web app or execution recording).
6. Submit completed work through the official submission form once released on **16 October 2026**.
7. Follow the submission instructions carefully and provide verified, publicly accessible links.
8. Submit original work, respect software licenses, and never publish passwords, API keys, or secrets.
9. **No payment is required at any stage of this internship.** CodeNova never charges fees.

⚠ IMPORTANT NOTE • MANDATORY COMPLETION CRITERIA

- **Minimum Task Requirement:** Interns must complete **at least 2 out of the 4** assigned tasks below to meet the minimum task-completion requirement.
- **Incomplete Threshold:** Completing only one task is strictly considered **incomplete** and will not qualify for certification.
- **Certificate Eligibility:** Certificate eligibility depends strictly on meeting the completion requirements and submitting acceptable work.
- **Deliverable Standard:** Each task submission must include source code on GitHub, a descriptive README.md, and a demo or screenshots where applicable.
- **Deadline Strictness:** Tasks must be submitted before the final deadline: **06 November 2026 (23:59 IST)**.
- **Zero Paid Tooling:** Do not require paid software, paid APIs, subscriptions, or paid hosting. Offer free or local alternatives (e.g., SQLite, free public APIs, LocalStack, GitHub Pages).

Domain Tasks — AI/ML (Part 1 of 2)

TASK 1 Exploratory Data Analysis

STATISTICAL EXPLORATION & VISUALIZATION

Objective & Scope: Clean a public dataset, create visualizations, and present meaningful observations.

CORE REQUIREMENTS & FEATURES

- Dataset ingestion and summary profiling (shape, data types, missing value analysis).
- Data cleaning: treatment of missing values, duplicate removal, and outlier detection.
- Visualizations: correlation heatmaps, feature distributions, boxplots, and pairwise scatter plots.
- Actionable narrative takeaways documented beneath each visualization explaining practical implications.

DELIVERABLES & ACCEPTANCE

RECOMMENDED STACK / TOOLS:

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

DELIVERABLES REQUIRED:

- Jupyter Notebook (.ipynb) with fully executed outputs and markdown commentary.
- Exported high-resolution visualization images.
- README.md summarizing key analytical insights and data characteristics.

ACCEPTANCE CRITERIA:

- Visualizations feature clear titles, axis labels, legends, and readable scales.
- Cleaning decisions (imputation vs. dropping) are thoroughly justified mathematically.
- Observations logically derive from empirical figures rather than generic speculation.

TASK 2 Spam Message Classifier

NATURAL LANGUAGE PROCESSING (NLP)

Objective & Scope: Train a text classifier, evaluate it with precision, recall, and F1-score, and demonstrate sample predictions.

CORE REQUIREMENTS & FEATURES

- Text preprocessing pipeline: tokenization, lowercasing, stop-word removal, and TF-IDF / Count vectorization.
- Model training comparing algorithms (e.g., Multinomial Naive Bayes vs. Logistic Regression).
- Rigorous model evaluation: Precision, Recall, F1-Score, and Confusion Matrix visualization.
- Interactive prediction function allowing manual testing with custom user-input messages.

DELIVERABLES & ACCEPTANCE

RECOMMENDED STACK / TOOLS:

Python, Scikit-Learn, NLTK or Regex, Pandas, Matplotlib / Seaborn

DELIVERABLES REQUIRED:

- Complete model training script or notebook.
- Saved vectorizer and model pipeline (pickle/joblib) or reproducible script.
- README.md reporting classification metrics table and sample prediction logs.

ACCEPTANCE CRITERIA:

- Achieves high precision to minimize false positives (marking legitimate messages as spam).
- Evaluation performed strictly on held-out test data without data leakage.
- Interactive test function accurately demonstrates real-time inference on new inputs.

Domain Tasks — AI/ML (Part 2 of 2)

TASK 3 Prediction Model

REGRESSION / CLASSIFICATION MODELING

Objective & Scope: Build a regression or classification model using a suitable public dataset, evaluate performance, and explain limitations.

CORE REQUIREMENTS & FEATURES

- Feature engineering, categorical encoding, and feature scaling.
- Train-test splitting and cross-validation to prevent overfitting.
- Model training using algorithms such as Random Forest, Gradient Boosting, or Linear/Ridge regression.
- Thorough evaluation using domain-appropriate metrics (RMSE/MAE for regression, ROC-AUC/F1 for classification) and limitation analysis.

DELIVERABLES & ACCEPTANCE

RECOMMENDED STACK / TOOLS:

Python, Scikit-Learn, Pandas, NumPy, Matplotlib

DELIVERABLES REQUIRED:

- Model training and evaluation codebase.
- Evaluation charts (residual plots, feature importance bar charts, or ROC curves).
- README.md discussing methodology, performance benchmarks, and known limitations.

ACCEPTANCE CRITERIA:

- Clear comparison against a baseline model demonstrating performance gains.
- Detailed discussion of model limitations, failure cases, and assumptions.
- Zero target leakage during feature engineering and preprocessing.

TASK 4 Document Question-Answering Assistant

RETRIEVAL & GROUNDED Q&A

Objective & Scope: Create an assistant that answers questions from provided documents, shows supporting source passages, and handles unsupported questions transparently.

CORE REQUIREMENTS & FEATURES

- Document text ingestion and semantic chunking.
- Information retrieval using TF-IDF / BM25 or free local embeddings (Sentence-Transformers / HuggingFace free models).
- Retrieval of top relevant passages matching user query.
- Grounded answer generation or extractive matching citing source passages, with transparent fallback when unsupported.

DELIVERABLES & ACCEPTANCE

RECOMMENDED STACK / TOOLS:

Python, Scikit-Learn / HuggingFace Transformers (free local models) or LangChain local, NLTK

DELIVERABLES REQUIRED:

- Question-answering assistant source code.
- Sample text document corpus used for testing.
- README.md with transcripts demonstrating grounded answers, passage citations, and unsupported queries.

ACCEPTANCE CRITERIA:

- Every answer explicitly displays the supporting source passage from the document.
- Transparently states 'Information not found in provided document' for ungrounded queries (zero hallucinations).
- Operates 100% locally or using free open-source tooling without paid API dependencies.

Internship Timeline & Key Milestones

PROGRAM MILESTONE	TARGET DATE	DESCRIPTION & ACTION REQUIRED
Project Assignment Date	10 October 2026	Official release of task specifications; interns commence project development.
Submission Form Release	16 October 2026	Official Google Form opens for early task submissions and project link uploads.
Mid-Program Progress Check	25 October 2026	Recommended self-paced checkpoint: completion of at least 1-2 practical tasks.
Final Submission Deadline	**06 November 2026**	Final cutoff at 23:59 IST. Submission portal closes. No late entries accepted.
Evaluation & Certification	Post-Deadline	Code verification, project evaluation, and distribution of Certificates & LORs.

Submission Details & Guidelines

When you have completed at least 2 of the assigned tasks, submit your work through the official CodeNova submission portal. Ensure your GitHub repositories are set to **Public**, your README files contain comprehensive documentation, and all deployment or demonstration links are active and functional.

OFFICIAL SUBMISSION FORM RELEASE

The official submission form link will be provided on 16 October 2026 via email & official community channels.

Releases 16 Oct 2026

Pre-Submission Quality Checklist

- ✔ Completed at least 2 out of the 4 assigned tasks
- ✔ README.md contains features, tech stack, and setup steps
- ✔ Zero API keys, passwords, or credentials in git history
- ✔ GitHub repository visibility set to Public
- ✔ Execution screenshots or live demo links included
- ✔ LinkedIn update shared tagging official CodeNova page

Completely Free Educational Initiative: CodeNova operates on a 100% free educational model. No registration fees, onboarding fees, certificate fees, or other charges are ever collected. If anyone requests payment claiming to represent CodeNova, report it immediately to **codenova.community@gmail.com**.

Official Contact Information & Community Links

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OFFICIAL LINKEDIN PAGE
CodeNova Community

Welcome to the CodeNova Community: We are thrilled to have you join Batch 01 of the CodeNova Internship Program. Our mission is to empower students worldwide to learn by doing, build real software, and grow into confident engineering professionals. Approach each task with curiosity, write clean code, help your peers, and build an exceptional portfolio!

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